

```

; Problem Statement: Write X86 ALP to find:
; a) Number of Blank spaces
; b) Number of lines
; c) Occurrence of a particular character.
; Accept the data from the text file. The text file has to be
; accessed during Program_1 execution and write FAR
; PROCEDURES in Program_2 for the rest of the processing.
; Use of PUBLIC and EXTERN directives is mandatory.

extern far_proc      ; External FAR procedure for text
processing (defined elsewhere)

global filehandle, char, buf, abuf_len

%include "macro2.asm" ; Include macro file (assumed to
define macros like fopen, fread, etc.)

;
*****
****
section .data
    nline db 10      ; Newline character (for line counting)
    nline_len equ $ - nline ; Calculate the length of newline
message

    ; Assignment header message
    ano db 10, "Write X86 ALP to find, a) Number of Blank
spaces b) Number of lines c) Occurrence of a particular
character. Accept the data from the text file. The text file has
to be accessed during Program_1 execution and write FAR
PROCEDURES in Program_2 for the rest of the processing.
Use of PUBLIC and EXTERN directives is mandatory", 10,
"Name:- Sumaanyu", 10, "Roll:- 7256", 10, "Date of
Performance:- 17/04/2025", 10
    db 10, "-----", 10
    ano_len equ $ - ano ; Calculate the length of the
assignment header

    filemsg db 10, "Enter filename for string operation : " ;
Prompt for filename
    filemsg_len equ $ - filemsg

    charmsg db 10, "Enter character to search : " ;
Prompt for character to search
    charmsg_len equ $ - charmsg

    errmsg db 10, "ERROR in opening File...", 10 ; Error
message for file opening failure
    errmsg_len equ $ - errmsg

    exitmsg db 10, 10, "Exit from program...", 10, 10 ; Exit
message
    exitmsg_len equ $ - exitmsg

;
*****
****
section .bss
    buf resb 4096 ; Buffer to hold the contents of the file
    buf_len equ $ - buf ; Buffer length

```

```

filename resb 50 ; Buffer for filename input
char resb 2 ; Buffer for character input

filehandle resq 1 ; Store the file handle (pointer to file)
abuf_len resq 1 ; Actual length of data read into buffer

;
*****
****
section .text
    global _start

_start:
    ; Display the assignment information
    display ano, ano_len

    ; Prompt the user for a filename
    display filemsg, filemsg_len
    accept filename, 50
    dec rax
    mov byte [filename + rax], 0 ; Null-terminate the filename

    ; Prompt the user for the character to search
    display charmsg, charmsg_len
    accept char, 2

    ; Open the file specified by the user
    fopen filename
    cmp rax, -1h ; Check if file opening failed (returns -1
on error)
    jle Error ; Jump to error handling if file open fails
    mov [filehandle], rax ; Store file handle

    ; Read the contents of the file into the buffer
    fread [filehandle], buf, buf_len
    mov [abuf_len], rax ; Store the actual length of data read

    ; Call the FAR procedure to process the file contents
(count blanks, lines, occurrences of character)
    call far_proc

    ; Exit the program
    jmp Exit

Error:
    ; Display the error message if file opening fails
    display errmsg, errmsg_len

Exit:
    ; Display the exit message and exit the program
    display exitmsg, exitmsg_len
    display nline, nline_len ; Add a newline at the end of output

    ; Exit syscall
    mov rax, 60 ; Syscall: exit
    mov rdi, 0 ; Exit status: 0
    syscall

```